

Surgery to the lymph nodes for breast cancer

There are different types of lymph node surgery. Your surgeon may remove some or all the lymph nodes in your armpit to check them for invasive breast cancer cells.

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What is surgery to the lymph nodes?

Your surgeon may remove some or all the [lymph nodes](#) in your armpit to check them for cancer cells. This can also:

- remove any lymph nodes that contain cancer cells
- give information about the stage of the cancer.

There are different types of lymph node surgery. You may need more treatment if only some lymph nodes were removed.

Sentinel lymph node biopsy (SLNB)

A sentinel lymph node biopsy (SLNB) is a way of checking the lymph nodes in the armpit. The lymph nodes most likely to have cancer cells are called the sentinel lymph nodes. These are the first ones that lymph fluid drains to from the breast area.

Your surgeon or nurse will explain if an SLNB is an option for you.

You will have an SLNB if the [ultrasound](#) or [fine needle aspiration](#) of your armpit was normal.

During an SLNB, surgeons remove the smallest number of lymph nodes possible (usually 1 to 3). This reduces the risk of side effects that can happen after lymph node surgery. These include swelling ([lymphoedema](#)) of the arm and stiffness of the arm and shoulder.

If there are no cancer cells in the sentinel nodes, you will not need any further treatment to the lymph nodes.

If cancer cells are found in the sentinel nodes, your cancer doctor will talk to you about whether you need further treatment. You may be offered another operation to remove the remaining lymph nodes. Some people have [radiotherapy](#) to the remaining lymph nodes instead of more surgery.

In some hospitals, the sentinel nodes can be checked for cancer cells during the operation to remove the cancer. This means that if more lymph nodes need to be removed, it can be done during the same operation.

Having an SLNB

Before the sentinel lymph nodes can be removed, the surgeon needs to check which nodes are the sentinel lymph nodes.

Before the surgery, the surgeon injects a harmless amount of radioactive liquid into your breast. During the operation, they use a handheld machine to find the lymph nodes that have picked up the radioactive liquid. The surgeon may also inject a blue dye into your breast during the operation. This stains the sentinel lymph nodes blue. The lymph nodes that pick up the radioactive liquid or become blue first are the sentinel lymph nodes.

The surgeon can then remove the blue or radioactive nodes (sentinel nodes). These are tested to see if there are any cancer cells in them.

Other ways to find sentinel nodes

Some people have their sentinel nodes detected using a magnetic tracer. The magnetic tracer is a liquid containing tiny iron particles. It is injected into the breast tissue, where it is absorbed by the lymph vessels and sentinel nodes. A doctor or nurse can then use a magnetic probe to detect the sentinel nodes. Your doctor or nurse can explain more about magnetic tracers.

Removing all the lymph nodes

Sometimes the surgeon will recommend removing all the lymph nodes in the armpit. This is called an axillary lymph node dissection (ALND) or axillary lymph node clearance (ANC). It aims to remove any nodes that contain cancer cells in the area close to the cancer.

You usually have an ALND if:

- a [fine needle aspiration](#) or biopsy of the lymph nodes finds cancer cells
- an SLNB or sample of the lymph nodes shows cancer cells in the lymph nodes.

An ALND increases the risk of swelling of the arm. This is called lymphoedema. We have more information about [lymphoedema and taking care of your arm](#).

About our information

References



Below is a sample of the sources used in our breast cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

ESMO. Early breast cancer clinical practice guidelines for diagnosis, treatment and follow-up. 2019, Vol 30, pp1192–1220. Available from: <https://www.esmo.org/guidelines/guidelines-by-topic/breast-cancer/early-breast-cancer> [accessed 2023].

National Institute for Health and Care Excellence (NICE). Early and locally advanced breast cancer: diagnosis and management. 2018. Updated 2023. Available from: <https://www.nice.org.uk/guidance/ng101> [accessed 2023].

Reviewers

This information has been written, revised and edited by Macmillan Cancer Support’s Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by [Dr Rebecca Roylance](#), Consultant Medical Oncologist and [Professor Mike Dixon](#), Professor of Surgery and Consultant Breast Surgeon.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF’s 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

Reviewed: 01 October 2023 | Next review: 01 October 2026

ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

[Get help](#)